

Supplementary material

Red blood cell (RBC) suspensions in confined microflows: pressure-flow relationship

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Device fabrication

Two 4" master wafers were fabricated. The first wafer was made using standard SU-8 photolithography and served as a template for creating the straight channel (see main article, Fig. 1a,c). A clean 4" silicon wafer (SVM) was used without further cleaning or drying. SU-8 2025 photoresist (Microchem) was spin coated on the wafer at 500 rpm for 10 s with a 100 rpm/s ramp followed by 1400 rpm for 30 s with a 1500 rpm/s ramp to a height of ~ 50 μm . The wafer was then soft-baked at 65°C for 5 min and at 95°C for 25 min on two separate hot plates. UV exposure was performed using an MA6 mask aligner (SUSS) with an exposure energy of 170 mJ·cm⁻². After a post exposure bake at 95°C for 5 min on a hot plate, the wafer was developed in 1-methoxy-2-propanol acetate (PGMEA) for 5 min, washed with isopropanol and spin dried. Details on the master fabrication of the confined network are discussed in previous work [1].

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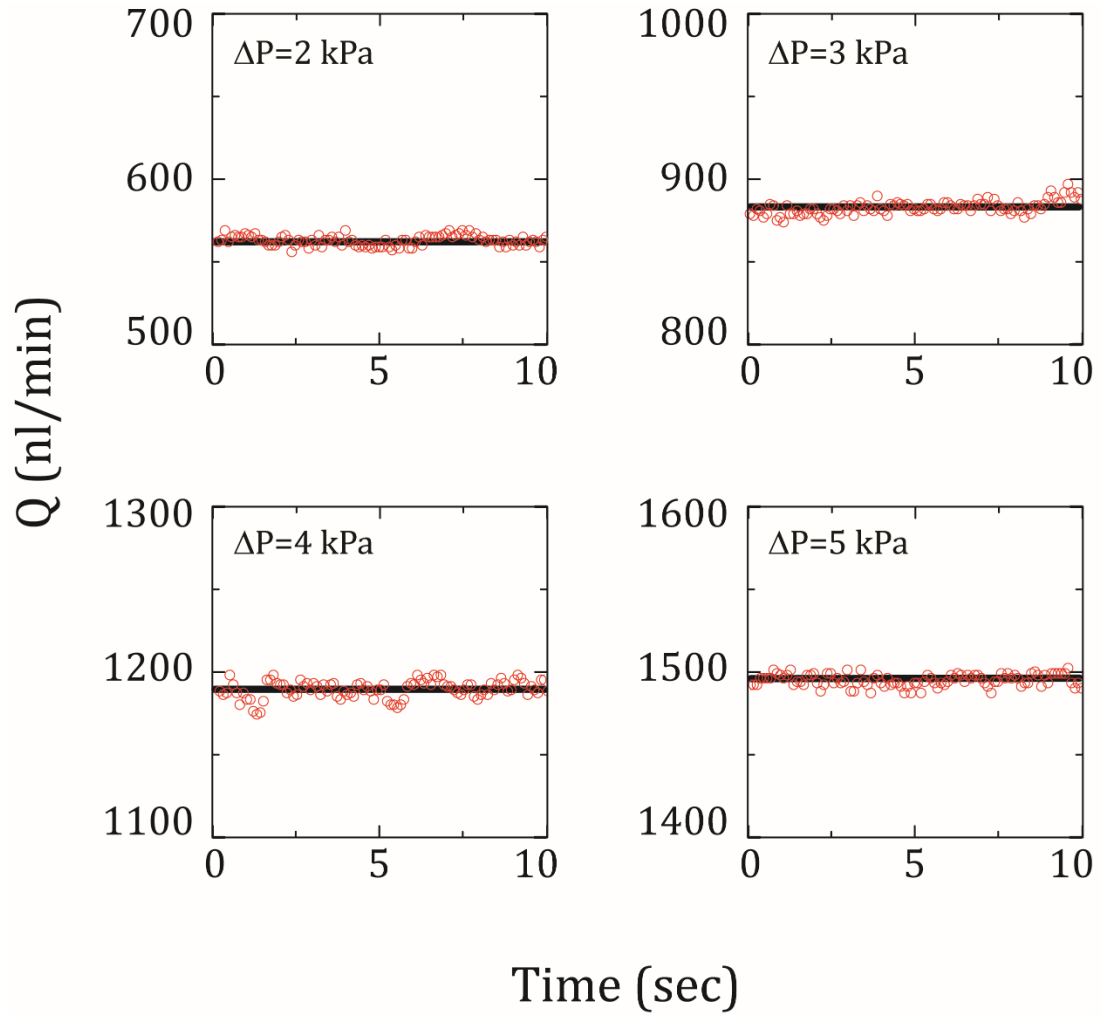


Figure S1: Examples of raw data measurements (Small Flow Unit, Fluigent SA, France) of flow rate (nl/min) versus sample time intervals for four distinct pressure settings (ΔP) in the microfluidic straight channel (see Fig. 1a in main manuscript) at 0% Hct. Namely, we measure 562.8 $\pm$ 3.5 nl/min ($\Delta P=2$ kPa), 883.2 $\pm$ 4.1 nl/min ($\Delta P=3$ kPa), 1190.2 $\pm$ 5.3 nl/min ($\Delta P=4$ kPa) and 1495.8 $\pm$ 3.6 nl/min ($\Delta P=5$ kPa), where values correspond to mean $\pm$ SD, such that SD is well below 5% of the mean.

Supplementary Material Video 1:

RBC suspension diluted in Dex-40 (Hct=23%) flowing in a microfluidic straight channel (SC) with a square cross section ($w \times h$) of 50 $\mu\text{m} \times 50 \mu\text{m}$ ($\Delta P=0.2$ kPa). The movie is played in real time.

Supplementary Material Video 2:

RBC suspension diluted in PBS (Hct=23%) flowing in a microfluidic straight channel (SC) with a square cross section ($w \times h$) of 50 $\mu\text{m} \times 50 \mu\text{m}$ ($\Delta P=0.2$ kPa). The movie is played x6 times slower than real time.

Supplementary Material Video 3:

RBC suspension diluted in Dex-40 (Hct=23%) flowing in a microfluidic network array ($\Delta P=2$ kPa). The movie is played x60 times slower than real time.

Supplementary Material Video 4:

RBC suspension diluted in PBS (Hct=23%) flowing in the microfluidic network array ($\Delta P=2$ kPa). The movie is played x60 times slower than real time.

References

[1] Stauber H, Waisman D, Korin N, Sznitman J. Red blood cell dynamics in biomimetic microfluidic networks of pulmonary alveolar capillaries. *Biomicrofluidics* 2017;11. doi:10.1063/1.4973930.